Problem Set 1
Due Date: Monday November 9th

Please submit your answers at the start of class. Physical copies are preferred, but digital submissions are acceptable as well. Please show your work, that is to say, be clear about which derivatives you are taking, how terms cancel in the algebra, etc.

Problem 1: CES: Profit Maximization

Consider a continuum of differentiated varieties indexed by $\omega \in \Omega$. The representative consumer has preferences

$$Q = \left[\int_{\Omega} q(\omega)^{\frac{\sigma-1}{\sigma}}, d\omega \right]^{\frac{\sigma}{\sigma-1}},$$

and faces prices $p(\omega)$ and total expenditure E.

(1) Set up and solve the consumer's problem of maximizing Q subject to the relevant budget constraint. Derive the first order condition, and show that for two varieties ω and ω', the following relationship holds:

$$\frac{q(\omega)}{q(\omega')} = \left(\frac{p(\omega)}{p(\omega')} \right)^{-\sigma}$$

(2) Derive the price index P such that $E = P \times Q$, and show that individual demand for variety ω satisfies

$$q(\omega) = \left(\frac{p(\omega)}{P} \right)^{-\sigma} \frac{E}{P}$$

(3) Suppose a monopolistically competitive firm with productivity φ faces CES demand over its output. Show that the firm's revenue is

$$r(\varphi) = EP^{\sigma-1} \left(\frac{\sigma}{\sigma-1} \frac{w}{\varphi} \right)^{1-\sigma},$$

and that the firm's profits are $\frac{r(\varphi)}{\sigma}$.

(4) Interpret the following cases. That is to say, describe consumer demand, firm markups, profits, and so on.

(a) $\sigma \to \infty$.

(b) $\sigma \to 0$.

(c) $\sigma \to 1$.

Problem 2: Chaney (2008)

Consider a static version of the Melitz model. Firms draw productivity φ from a Pareto distribution with lower bound $\varphi_{\min} > 0$ and shape parameter k. That is to say, the distribution satisfies

$$G(\varphi) = 1 - \left(\frac{\varphi_{\min}}{\varphi} \right)^{k}, \qquad \varphi \geq \varphi_{\min}.$$

After drawing a productivity, firms choose whether to exit, produce for the domestic market only, or produce for both the domestic market and for the export market. Domestic production incurs a fixed cost f_d. Exporting incurs an additional fixed cost f_x, and an iceberg trade cost τ. The domestic market has expenditure E and price index P, and the foreign market has expenditure $E^\star$ and price index $P^\star$.

(1) Derive the density $g(\varphi)$ and the probability that a productivity draw exceeds an arbitrary cutoff φ_c.

(2) Derive the distribution and density of productivity conditional on surviving the domestic cutoff φ_d^*.

(3) Calculate the average productivity of surviving firms, $E[\varphi \mid \varphi \geq \varphi_d^*]$.

(4) Let us now add CES demand on the consumer side, and assume that $k > \sigma - 1$. Normalize wages to be equal to 1. We know that in this case, revenue and profits are proportional to $\varphi^{\sigma-1}$. For an arbitrary cutoff φ_c, compute expected revenue conditional on exceeding the cutoff φ_c. What happens when $k \to \sigma - 1$ from above?

(5) Define the zero-profit cutoff as in class. Solve for the domestic production cutoff $\varphi_d^\star$ and the export cutoff $\varphi_x^\star$ as a function of both domestic and foreign aggregates, trade costs, fixed costs, and other model parameters.

(6) Compute the fraction of domestically active firms that are exporters. Discuss how this expression changes with trade costs τ, foreign market size $E^\star$, and foreign prices $P^\star$.

Problem 3: Hopenhayn on the Computer

Consider the version of the Hopenhayn model discussed in class. We will take a little shortcut, and define $A = EP^{\sigma-1}$, this will save us from computing the price index repeatedly (notice: in the model presented in class, all terms involving E and P had the two together in this form).

From period to period, log productivity follows the following process:

$$\log \varphi' = (1-\rho)\mu + \rho \log \varphi + \epsilon',$$

where $\epsilon \sim N(0, \sigma_\epsilon^2)$. Suppose we take the following parameter values:

Variable	Interpretation	Value
σ	Elasticity of Substitution	4
β	Discount factor	0.95
w	Wage	1 [normalization]
c_f	Fixed operating cost	0.25
c_e	Sunk entry cost	0.25
ρ	Productivity persistence	0.9
σ_ϵ	Variance of TFP shocks	0.15
μ	Mean of TFP process	$\log 1.5$
$\bar{L}$	Labour endowment	1

Further assume that entrants draw productivity from a log-normal distribution with mean 0, variance 0.1. This underlying distribution is not so important (think about why!), but will ease computation.

(1) Discretize a reasonable grid of productivity, and plot the firm's operating profit as a function of productivity. For now, "normalize" $A = 1$ [this is not a normalization, really, as we have taken $w = 1$. We will solve for A later].

(2) For a given value of $A = 1$, use value function iteration to compute $V(\varphi)$.

(3) Compute the continuation value $W(\varphi)$, and identify the exit cutoff productivity.

(4) Define $R(A) = \beta \mathbb{E}_G[V(\varphi)] - wc_e$. Compute the root to solve for $R(A) = 0$, and argue why this pins down the equilibrium object A.

(5) Let $\mu(\varphi)$ denote the mass of active firms at each productivity state. Given a mass of entrants M, a stationary distribution satisfies the law of motion given in the slides, with $\mu'(\varphi) = \mu(\varphi)$ for every φ. Compute the stationary distribution given a mass of entrants M by iterating on the law of motion.

(6) Calculate variable labor demand $n(\varphi) = q(\varphi)/\varphi$ and include the labor used to pay fixed operating costs and entry costs to recover the equilibrium mass of entrants M.

(7) Plot the stationary firm distribution. Report the mass of active firms and mean productivity among active firms.

(8) Suppose that fixed operating costs increase from 0.25 to 0.3. Report percentage changes in the following variables, and provide economic intuition for the direction of the change.

   (a) The equilibrium object A.

   (b) The exit cutoff.

   (c) The mass of entrants and active firms.

   (d) Mean productivity among active firms.